

### 3.3 Assessment criteria

#### A: Summary of case study or physics-based visit

| Ref | Criterion                                                                                                                                                                                                          | Mark     |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| S1  | Carries out a visit <b>OR</b> uses library, consulting a <b>minimum of three</b> different sources of information (eg books/websites/journals/magazines/case study provided by Edexcel/manufacturers' data sheets) | 1        |
| S2  | States details of visit venue <b>OR</b> provides <b>full</b> details of sources of information                                                                                                                     | 1        |
| S3  | Provides a brief description of the visit <b>OR</b> case study                                                                                                                                                     | 1        |
| S4  | Makes correct statement on relevant physics principles                                                                                                                                                             | 1        |
| S5  | Uses relevant specialist terminology correctly                                                                                                                                                                     | 1        |
| S6  | Provides one piece of relevant information (eg data, graph, diagram) that is not mentioned in the briefing papers for the visit or case study                                                                      | 1        |
| S7  | Briefly discusses context (eg social/environmental/historical)                                                                                                                                                     | 1        |
| S8  | Comments on implication of physics (eg benefits/risks)                                                                                                                                                             | 1        |
| S9  | Explains how the practical relates to the visit or case study                                                                                                                                                      | 1        |
|     | <b>Maximum marks for this section</b>                                                                                                                                                                              | <b>9</b> |

## B: Planning

| Ref | Criterion                                                                                                                                                         | Mark      |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| P1  | Lists all material required                                                                                                                                       | 1         |
| P2  | States how to measure one relevant quantity using the most appropriate instrument                                                                                 | 1         |
| P3  | Explains the choice of the measuring instrument with reference to the scale of the instrument as appropriate and/or the number of measurements to be taken        | 1         |
| P4  | States how to measure a second relevant quantity using the most appropriate instrument                                                                            | 1         |
| P5  | Explains the choice of the second measuring instrument with reference to the scale of the instrument as appropriate and/or the number of measurements to be taken | 1         |
| P6  | Demonstrates knowledge of correct measuring techniques                                                                                                            | 1         |
| P7  | States which is the independent and which is the dependent variable                                                                                               | 1         |
| P8  | Identifies and states how to control all other relevant variables to make it a fair test                                                                          | 1         |
| P9  | Comments on whether repeat readings are appropriate in this case                                                                                                  | 1         |
| P10 | Comments on safety                                                                                                                                                | 1         |
| P11 | Discusses how the data collected will be used                                                                                                                     | 1         |
| P12 | Identifies the main sources of uncertainty and/or systematic error                                                                                                | 1         |
| P13 | Draws an appropriately labelled diagram of the apparatus to be used                                                                                               | 1         |
| P14 | Plan is well organised and methodical, using an appropriately sequenced step-by-step procedure                                                                    | 1         |
|     | <b>Maximum marks for this section</b>                                                                                                                             | <b>14</b> |

## C: Implementation and Measurements

| Ref | Criterion                                                                                                           | Mark     |
|-----|---------------------------------------------------------------------------------------------------------------------|----------|
| M1  | Records all measurements using the correct number of significant figures, tabulating measurements where appropriate | 1        |
| M2  | Uses correct units <b>throughout</b>                                                                                | 1        |
| M3  | Obtains an appropriate number of measurements                                                                       | 1        |
| M4  | Obtains measurements over an appropriate range                                                                      | 1        |
|     | <b>Maximum marks for this section</b>                                                                               | <b>4</b> |

## D: Analysis

| Ref                            | Criterion                                                                | Mark |
|--------------------------------|--------------------------------------------------------------------------|------|
| A1                             | Produces a graph with appropriately labelled axes and with correct units | 1    |
| A2                             | Produces a graph with sensible scales                                    | 1    |
| A3                             | Plots points accurately                                                  | 1    |
| A4                             | Draws line of best fit (either a straight line or a smooth curve)        | 1    |
| A5                             | Comments on the trend/pattern obtained                                   | 1    |
| A6                             | Derives relation between two variables or determines constant            | 1    |
| A7                             | Discusses/uses related physics principles                                | 1    |
| A8                             | Attempts to qualitatively consider sources of error                      | 1    |
| A9                             | Suggests realistic modifications to reduce error/improve experiment      | 1    |
| A10                            | Calculates uncertainties                                                 | 1    |
| A11                            | Provides a final conclusion                                              | 1    |
| Maximum marks for this section |                                                                          | 11   |

## E: Report

| Ref                            | Criterion                                           | Mark |
|--------------------------------|-----------------------------------------------------|------|
| R1                             | Summary contains few grammatical or spelling errors | 1    |
| R2                             | Summary is structured using appropriate subheadings | 1    |
| Maximum marks for this section |                                                     | 2    |

|                           |    |
|---------------------------|----|
| Total marks for this unit | 40 |
|---------------------------|----|